TIPA Korea Technology and Information Promotion Agency for SMEs

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	**Optical Sensing Technologies and Algorithms for Vital Sign Monitoring in Wearable Device**			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio (√), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI (√), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Bio · Medical		Electrical·Electronics	
	Medium Category	Small Category	Medium Category	Small Category
	Medical Information & Systems	Bio-signal Measurement / Diagnostic Devices	Optical Application Devices	Optical Measurement and Control Devices
Concept of Proposed Technology	The proposed technology is based on optical monitoring approaches covering a range of technology readiness levels, including high-TRL methods such as reflective photoplethysmography (PPG) as well as lower-TRL techniques such as laser Doppler flowmetry and speckle-based measurements, to enable the extraction of diverse biomarkers. Key aspects of systems developed with this technology include optical and mechanical design, precise analog front-end control, and advanced embedded algorithms. In this domain, CSEM has extensive experience and offers a broad portfolio of monitoring capabilities, ranging from fitness and activity tracking—such as heart rate, energy expenditure, and VO_2max—to innovative parameters for medical applications, including blood pressure monitoring, cardiac arrhythmia detection, and oxygen saturation, cardiac output, as well as wellness-oriented features such as sleep monitoring, stress assessment, and heart rate variability (HRV) analysis..			
Rationale for Joint R&D Project	CSEM has strong competences and globally recognized expertise in the development of systems based on optical monitoring to extract vital signs and digital biomarkers. However, to translate this technological excellence into real-world impact, it requires collaborations with industry and other actors. In the frame of a joint R&D project, Korean partners could offer complementary strengths in large-scale manufacturing, miniaturized electronics, and rapid consumer-device commercialization, in addition to opportunities for deployment of developed solutions in the Korean healthcare systems and clinical studies to demonstrate impact.			
Key Words	1.Optical sensing 2.Photoplethysmography (PPG) 3.Wearable sensors			

	4.Cardiovascular biomarkers 5.Physiological sensing

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level 6	End of the Project	Level 9
R&D Objectives	Through a project supported by this scheme, we aim to develop systems integrating novel optical sensing schemes, which could include methods such as laser doppler flowmetry or laser speckle imaging, exploring multi-wavelength and novel optomechanical configurations to enhance signal quality, robustness, and energy efficiency, as well as to enable new insights and biomarkers. Alternatively, the project can adapt and extend well-established optical monitoring technologies such as PPG in novel contexts, e.g. at new body locations, for previously not tackled clinical problems, or challenging daily life use cases where current solutions remain limited.			
R&D Target Specifications	Examples of possible targets: - Heart Rate accuracy ±3 bpm (rest), ±5 bpm (activity) - SpO_2 accuracy ±3% (70–100%) - Optical BP error ≤ ±5 mmHg (mean), SD ≤ 8 mmHg - Arrhythmia detection ≥95% sensitivity, ≥95% specificity - Sleep stage classification ≥75% agreement with reference - VO_2max estimation error ≤10% - Energy expenditure error ≤10–15%			
R&D Plan	Specific timelines depend on project scope and defined end goals. The typical duration for a development project for optical monitoring systems can be 1 to 3 years. The main stages of the project will be as follows: 1) **Requirements definition** – This stage will establish functional, technical, and clinical requirements to ensure the developed system will meet performance, usability, and regulatory expectations. 2) **Design of optomechanical and electronic subsystems** – Including selection of optical sensing scheme and its physical implementation, and the electronic component selection as well as layout design. 3) **Embedded software implementation** – Implementation and optimization of firmware and signal processing- algorithms to ensure reliable operation under diverse motion and environmental conditions. 4) **System-level integration** – Combination of all hardware and software components into a cohesive prototype/demonstrator. 5) **Verification and validation of the developed system** – The integrated system will be tested to confirm that it meets performance specifications for the targeted use cases. 6) **Demonstration of impact via clinical studies** – The technology will be evaluated in real world clinical or daily -life- environments, preferably in the primary target market, e.g. the Korean healthcare system. 7) **Technology transfer** – Finalized designs, documentation, and know-how will be transferred to industrial partners to enable further productization and commercialization.			

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	For optical monitoring systems, CSEM has the expertise to develop all aspects including hardware, software, signal processing, system-level integration, as well as data management and analysis capabilities on the cloud and edge. Depending on the strengths of the Korean partner and desired project outcomes, CSEM can take on a portion or the entirety of the development pipeline, including assembly and deployment of demonstrator units for the developed system. Alternatively, CSEM can also play a role in supporting the Korean partner in data collection and clinical studies to generate evidence of efficacy/impact for developed systems.
Expected R&R of Korean SME Partner	Depending on the project scope and objectives, the expected roles and responsibilities of the Korean partner may include the phases closely linked to system design and productization such as refinement of requirements, planning of product roadmaps, verification and validation activities, and preparation of data collection- studies. In cases where the Korean partner's expertise matches with the more technical WPs/tasks they could also have responsibilities there.

5. Expected Outcomes/Impact

Expected Outcomes and Impact	Potential and expected outcomes from such a project include the following: - Novel optical sensing schemes developed and their efficacy demonstrated in real-world clinical or daily use settings - Prototypes and demonstrators of wearable or other similar devices, reaching a technological maturity for industrialization and commercialization - (Potential long-term impact after further successful commercialization and clinical adaptation) reduced pressure on healthcare systems and improved population-wide health outcomes, first in the main countries of the collaboration (Korea and Switzerland), then worldwide, resulting from novel biomarker information (e.g. cardiovascular event warnings) to be used in early diagnosis/prevention

6. References

(1)

(2)

☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization